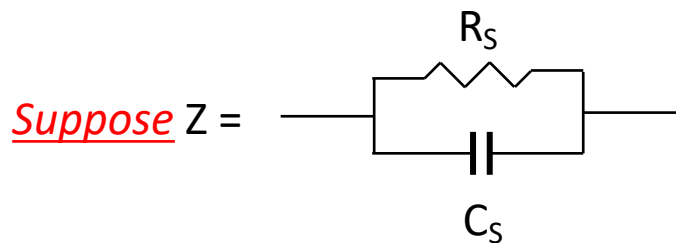
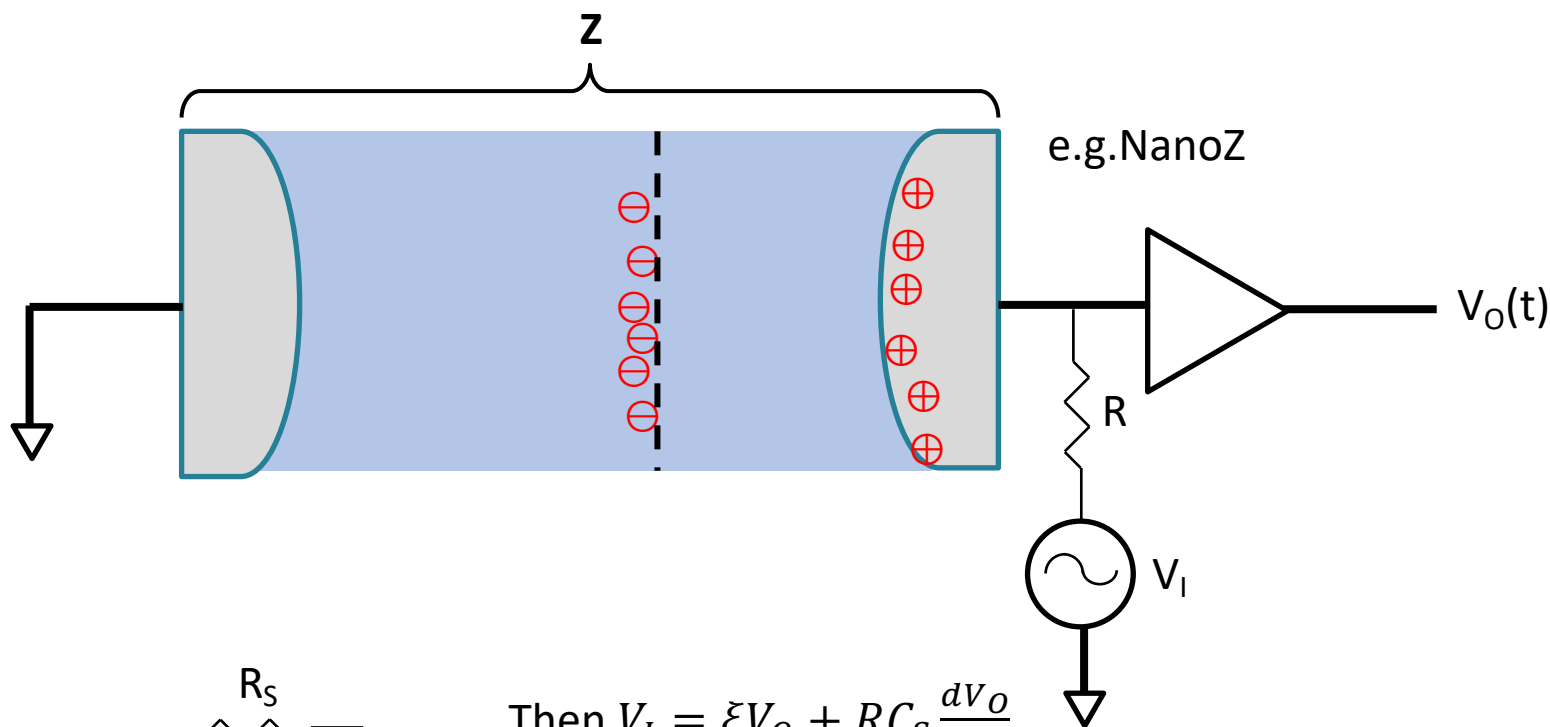


Impedance Measurement

What's a Neurotic to do to understand this?

Electrode Impedance Spectroscopy!

Oooh! Sounds so knowledgeable!



$$\text{Then } V_I = \xi V_O + RC_S \frac{dV_O}{dt}$$

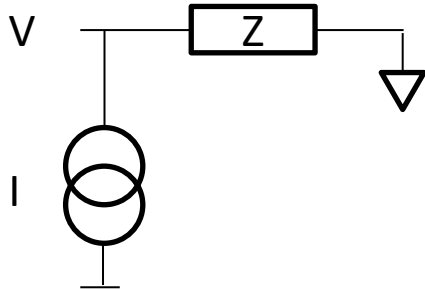
$$\text{Where } \xi = 1 + \frac{R}{R_S}$$

Impedance Measurement

If $V_I = \sin \omega t$ and $V_O = A \sin(\omega t + \varphi)$ then (with some arithmetic)

$$A = \frac{1}{\sqrt{\xi^2 + (\omega R C_S)^2}} \text{ and } \tan \varphi = -\frac{(\omega R C_S)}{\xi}$$

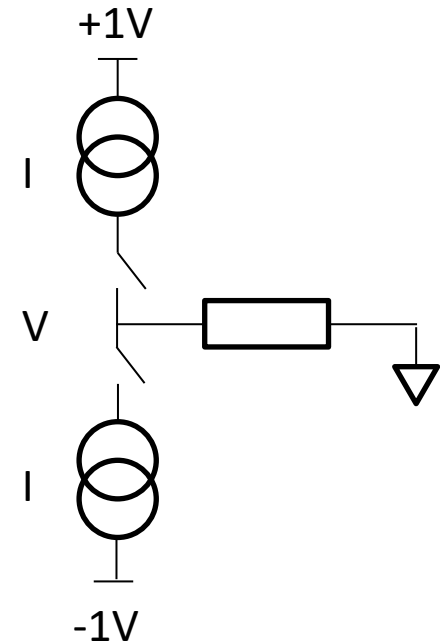
Note: this is only true for a “pure” resistor and capacitor



For EChip, instead of generating a voltage through a resistor, we generate a current (I) and measure the resulting voltage

Impedance Measurement with EChip

- General Idea: EChip can source or sink a current
 - $V = I \times R$, where $I \leq 250 \text{ nA}$
 - The sign of the current is determined by whether the **PLATE UP** or **PLATE DN** switch is turned on
- Set the GAIN corresponding to the estimate of R
 - e.g. if $R \sim 50 \text{ k}\Omega$ and $I_{\text{MAX}} = 250 \text{ nA}$ then $V_{\text{MAX}} = 12.5 \text{ mV} \rightarrow \text{GAIN} = 20$

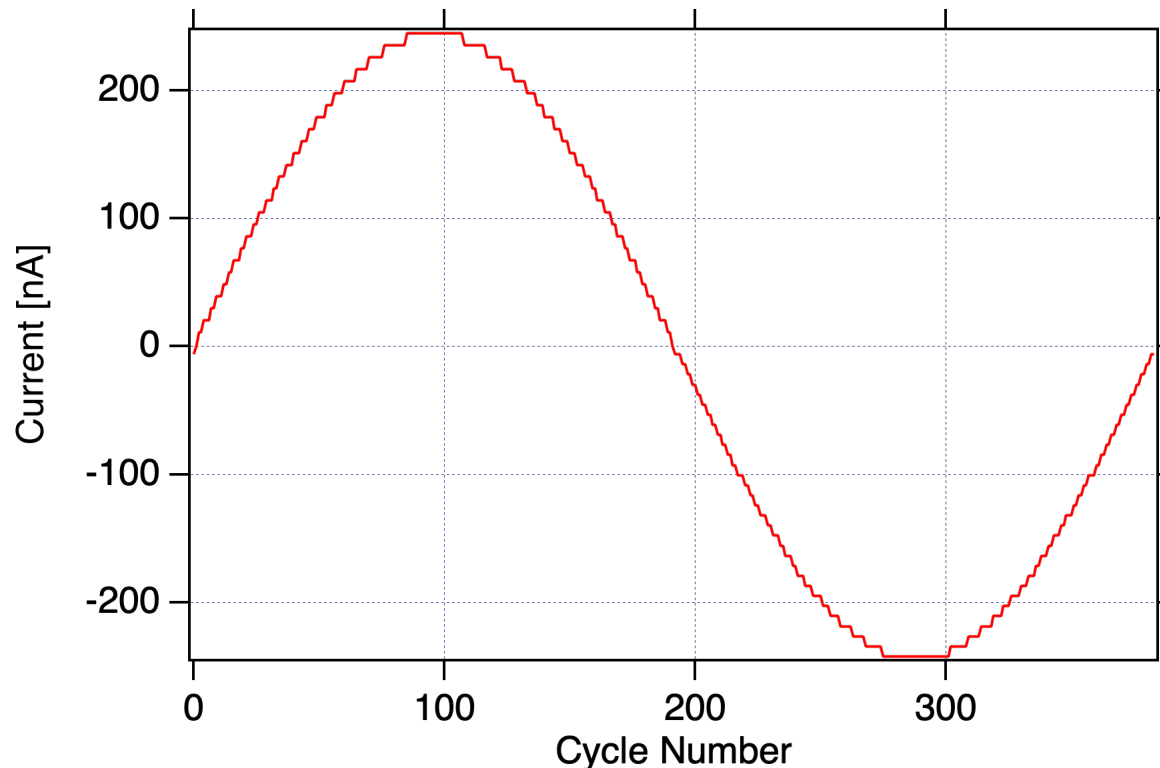


Specific Method

We want to measure the impedance of the pixel at Row R and Column C

1. Download ("register_write") basic configuration (RESET, DACs, GO)
2. Write the ROW register = R
3. Switch to FPGA-lookup table generated commands
4. Record...

The lookup table will (see next pages) generate a 1 kHz sine wave



Details – how this was done

HSCLK	8 ns
Serial Divider	8
Serial Clock	64 ns
Clocks/Word	20
Word Time	1280 ns
Words/ms	781.25
Round to	768
Program DAC	2 words
Cycles/2	384

Run the serial clock SCLK continuously

$SCLK = (\text{say}) HSCLK / 8$

Define a serial cycle as $20 \times SCLK$

For each cycle, there is a 20-bit word SDAT:

$SDAT[0..15] = \text{serial command } [0..15]$

$SDAT[16..19] = 0$

With the numbers above, 1 kHz is around 768 cycles

Lookup table has the following entries

Cycle	SDAT
0	COL register = desired pixel column
1	DATA register = gain, PLATE UP
2	WRITE
3	COL register = 0x44 (Plate DAC)
4	DATA register = sine value[0]
5	WRITE

Loop 192 times

Cycle	SDAT
389	COL register = desired pixel column
390	DATA register = gain, PLATE DOWN
391	WRITE
392	COL register = 0x44 (Plate DAC)
393	DATA register = sine value[0 + pi]
394	WRITE

Loop 192 times

Then back to cycle 0